

Host PC (Windows 11)

conversion and analysis only

WinMET dataset

five families, 100 samples each

Converter

call order preserved; repeats
expanded; encodings normalized

Analyzers (Python)

metrics, failure taxonomy,
Sysmon parsing, statistics

converted logs

PowerShell
Direct
(no network)

**results +
Sysmon events**

Isolated VM (Windows 10, Hyper-V)

all re-execution; all real side effects confined

For each of the 500 samples, in order:

1. clear the sandbox directory and the sandbox registry sub-tree
2. take the registry listing before the run
3. re-execute the converted log:
single-threaded, in recorded order
4. count files, registry entries and
non-loopback connections
outside the sandbox
5. dry run of the same sample;
wall-clock time recorded for both

Telemetry runs only (100 samples):

Sysmon v15.21 active, recording only
the tool's own events